

Homework (Carolina Reaper)

- Q1.** A painter is creating a mural on a 24×36 canvas. If she wants to divide it into smaller squares of equal size, what is the largest possible square size s in inches if she wants the number of squares to be a quadratic equation in the form $x^2 + bx + c$, where b and c are positive integers? The dimensions of the squares are $s \times s$.
- (A) 1×1
 (B) 2×2
 (C) 3×3
 (D) 4×4
 (E) 6×6
- Q2.** A musician is composing a piece with a rhythm pattern that follows a quadratic equation $x^2 - 5x + 6 = 0$. What are the possible time signatures that can be used for this pattern if the solutions to the equation represent the number of beats in a measure?
- (A) $2/4$ and $3/4$
 (B) $3/4$ and $2/4$
 (C) $4/4$ and $6/4$
 (D) $6/4$ and $4/4$
 (E) $2/4$ and $6/4$
- Q3.** A photographer is taking pictures of a ballet dancer. If the height h of the dancer above the ground at time t is modeled by the quadratic equation $h(t) = -2t^2 + 8t + 1$, at what time t will the dancer be at a height of 5 feet?
- (A) $t = 1$
 (B) $t = 2$
 (C) $t = 3$
 (D) $t = 4$
 (E) $t = 5$
- Q4.** A sound engineer is mixing a song with two different frequencies, f_1 and f_2 , where f_1 and f_2 are the solutions to the quadratic equation $x^2 - 7x - 18 = 0$. What are the two possible frequencies?
- (A) $f_1 = 9$ Hz, $f_2 = -2$ Hz
 (B) $f_1 = -2$ Hz, $f_2 = 9$ Hz
 (C) $f_1 = 2$ Hz, $f_2 = -9$ Hz
 (D) $f_1 = -9$ Hz, $f_2 = 2$ Hz
 (E) $f_1 = 18$ Hz, $f_2 = -1$ Hz
- Q5.** An artist is creating a mosaic with two different colors of tiles, where the total number of tiles t is related to the number of blue tiles b by the quadratic equation $t = b^2 + 4b + 4$. If $t = 36$, how many blue tiles are there?
- (A) $b = 2$
 (B) $b = 4$
 (C) $b = 6$
 (D) $b = 8$
 (E) $b = 10$
- Q6.** A choreographer is creating a dance routine that involves n dancers. The number of possible formations f is given by the quadratic equation $f = n^2 - 3n + 2$. If $f = 10$, how many dancers are there?
- (A) $n = 5$
 (B) $n = 6$
 (C) $n = 7$
 (D) $n = 8$
 (E) $n = 9$
- Q7.** A sculptor is creating a statue with a base that has a quadratic cross-section given by $x^2 + 5x + 6 = 0$. What are the possible dimensions x of the base?
- (A) $x = -2$ and $x = -3$
 (B) $x = -3$ and $x = -2$
 (C) $x = 2$ and $x = 3$
 (D) $x = -2$ and $x = 3$
 (E) $x = 2$ and $x = -3$
- Q8.** A music producer is mixing a song with a frequency response curve that follows a quadratic equation $y = x^2 - 2x - 3$. If the frequency x is in Hz and the amplitude y is in decibels, at what frequency will the amplitude be -4 decibels?
- (A) $x = 3$ Hz
 (B) $x = 4$ Hz
 (C) $x = 5$ Hz
 (D) $x = 6$ Hz
 (E) $x = 7$ Hz

Open Ended Questions (FRQ)

- Q9.** A photographer is taking a picture of a model on a staircase. The height h of the model above the ground at time t is given by the quadratic equation $h(t) = -4t^2 + 12t + 2$. Show that the solutions to the equation $h(t) = 6$ are $t = 1$ and $t = 3$. You must show all work and explain your reasoning. Assume t is in seconds.

- Q10.** An artist is creating a painting with a series of concentric circles. The radius r of each circle is related to the number of circles n by the quadratic equation $r = n^2 - 2n + 1$. Find the radius of the circle when $n = 4$. Show all work and explain your reasoning.

Chef's Tips

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- Tip 1: Check that all quadratic equations are in the standard form $ax^2 + bx + c$.
 - Tip 2: Verify all solutions by plugging them back into the original equation.
 - Tip 3: Make sure to consider the context of each problem and use it to inform your solution.